

The Evolution of NFL Offensive Strategy

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Background

- All of us have interests corresponding with professional sports.
- Of course, we wanted to cater our final project centering around sports statistics.
- NFLFastR scrapes public NFL play-by-play data into a reliable R package
- NFLFastR is an ideal foundation for studying how offensive strategy has evolved in the league

History

Our Mission

- Understand the changes in offensive strategy in the time period between 2000 to 2025
- Discern the effects of certain rule changes on NFL offense
- Present our findings to NFL Executives, Teams, and Analysts

Approach

- Use NFLFastR play-by-play data to build a consistent and reproducible dataset
- Define key offensive metrics for comparison across eras
- Create visualizations that highlight league-wide shifts in behavior rather than team-specific noise
- Interpret results in the context of rule changes and broader analytical trends in football

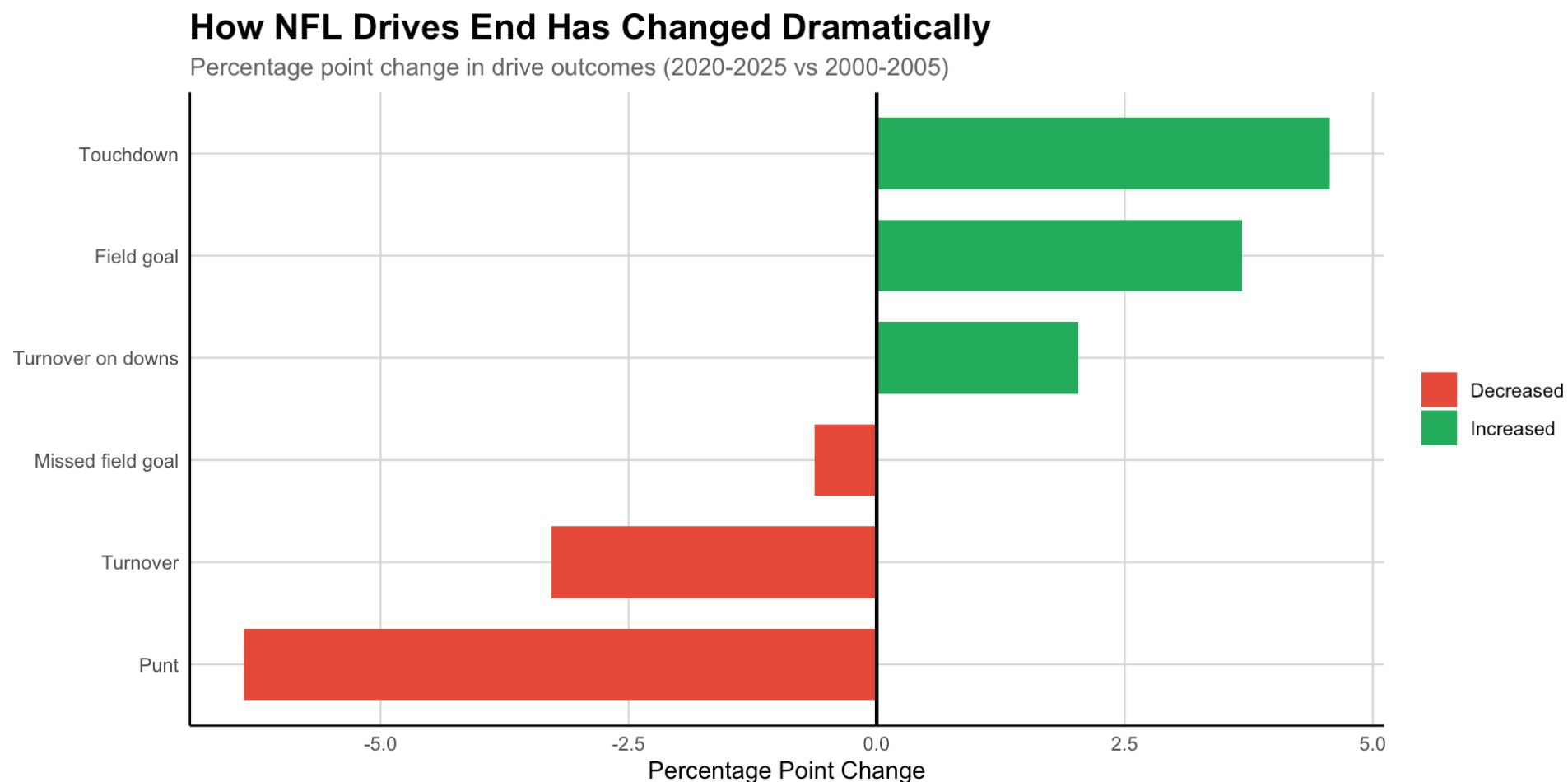
Generating Our SQLite

```
1 #| echo: true
2 years <- 2000:2025
3 dcon <- dbConnect(RSQLite::SQLite(), "nfl_pbp.sqlite")
4
5 # creates tables for specific years (2000-2025)
6 for(year in years){
7   message("Loading ", year, "...")
8   pbp <- nflfastR::load_pbp(year)
9   table_name <- paste0("pbp_", year)
10
11   dbWriteTable(dcon, table_name, pbp, overwrite = TRUE)
12
13   message("Created Table: ", table_name)
14 }
15
16 # creates the general play_by_play data from all years
17 all_queries <- paste0("SELECT * FROM pbp_", years)
18 union_query <- paste(all_queries, collapse = " UNION ALL ")
```

Offensive Conversion

- Drives now end in touchdowns and field goals more often than in the early 2000s
- Red zone execution have increased across teams
- Modern offenses waste fewer possessions through punts and turnovers

Changes in NFL Drives Between Eras (2000-2005 and 2020-2025)

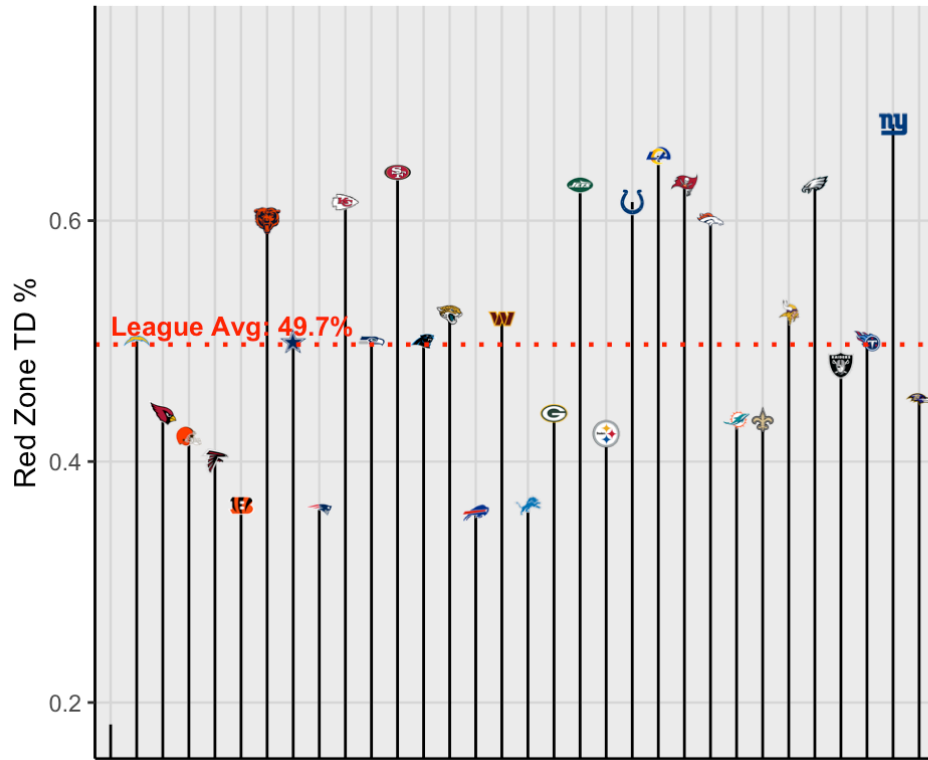


Comparing Red Zone Touchdown Efficiency

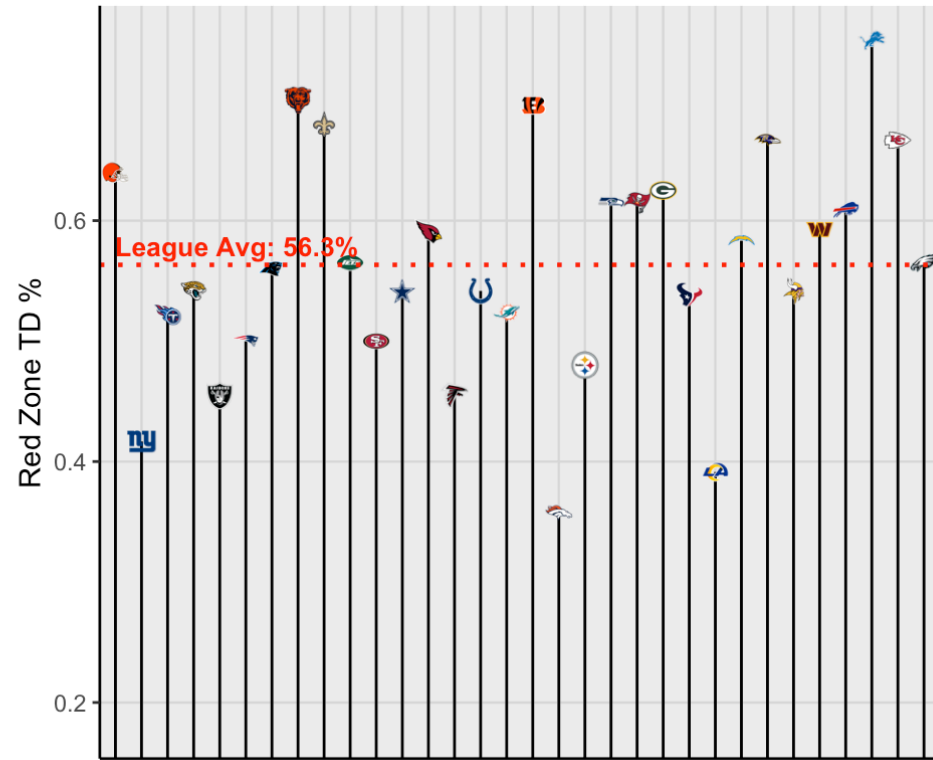
Red Zone Touchdown Efficiency

2000 vs. 2024 (Sorted by Wins)

2000 Season



2024 Season



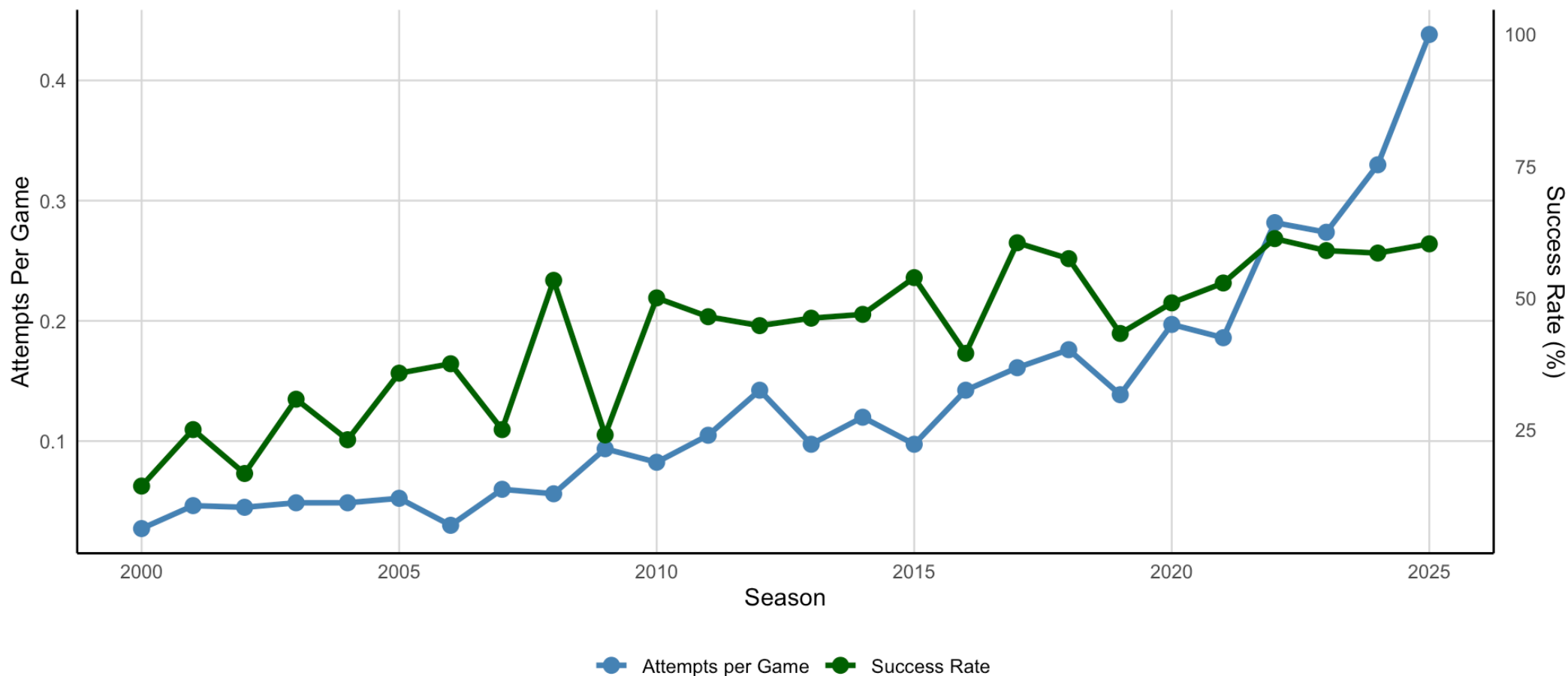
Kicking Throughout the NFL

- Field Goals have become a staple of the modern NFL offensive game
- Helps secure easier points when touchdowns aren't guaranteed
- Gradual evolution throughout the years
- Rule Changes have affected kicking instantaneously

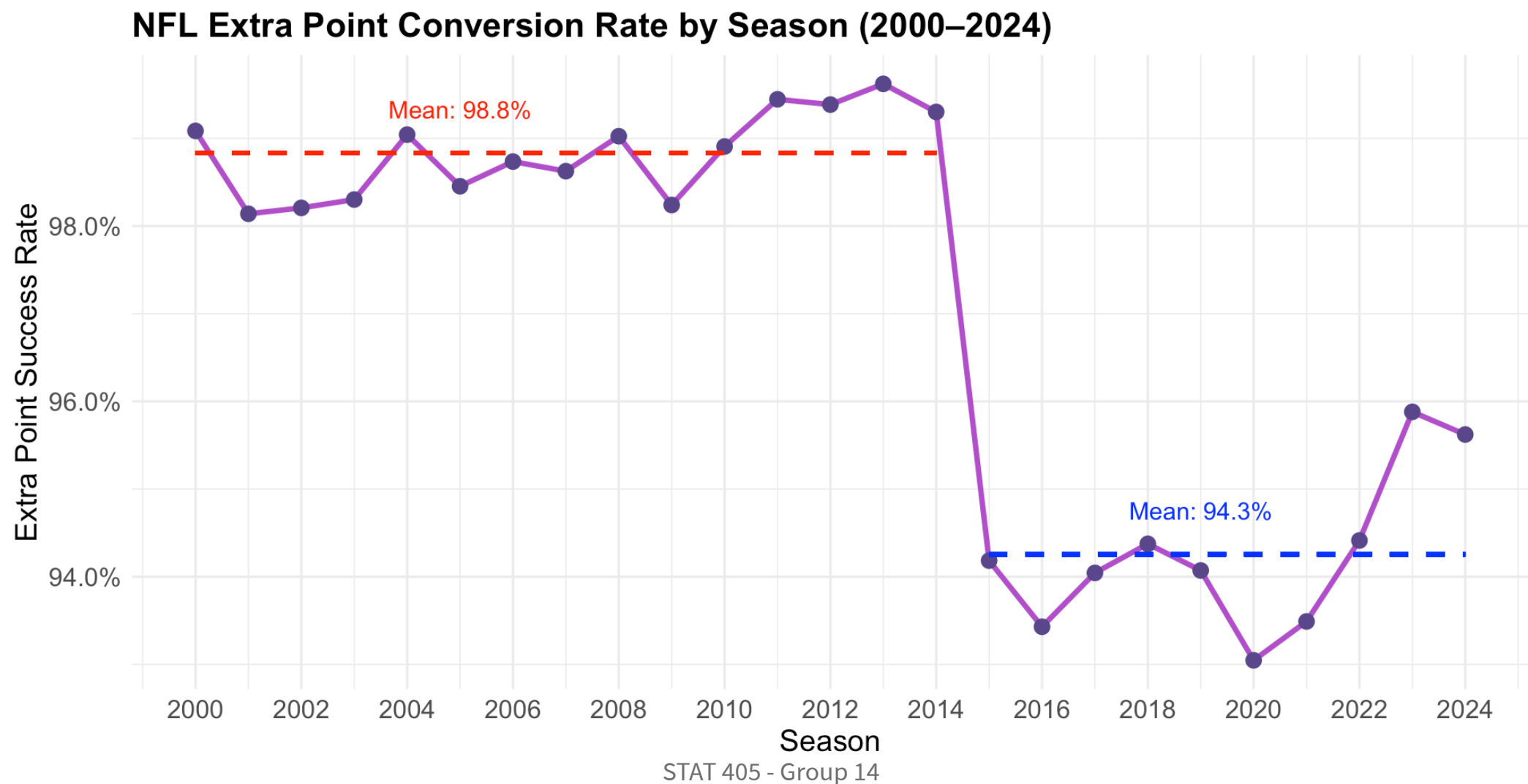
Upward Long Range Field Goal Trends

55+ Yard Field Goal Attempts & Success Rate

NFL Regular Season (2000-2025)



A Dip in the Extra Point Conversion Rate

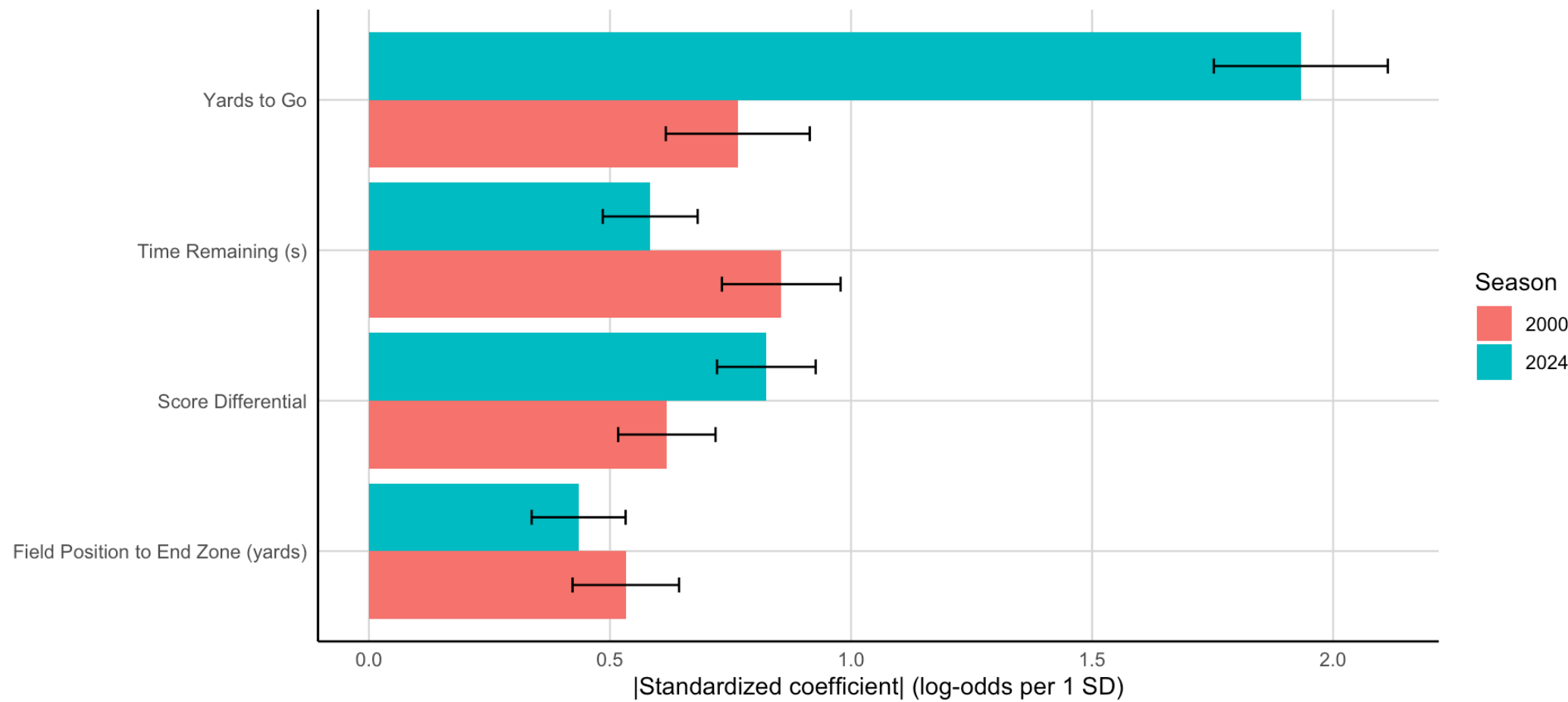


Late Down Decisions

- Teams have become more aggressive with fourth down decisions
- Efficiency and offense have evolved, even in clutch time

Crucial Plays in Crunch Time

Change in relative importance of 4th-down factors
(with 95% CIs)

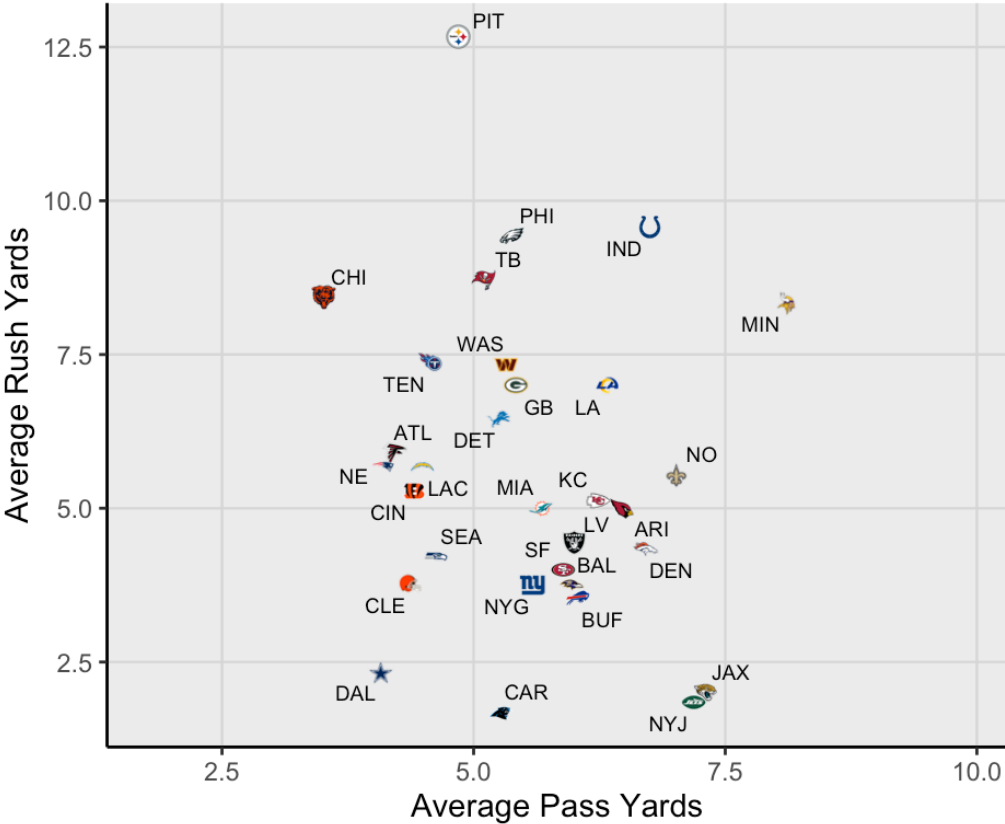


Rush to Pass Yards in Crunch Time

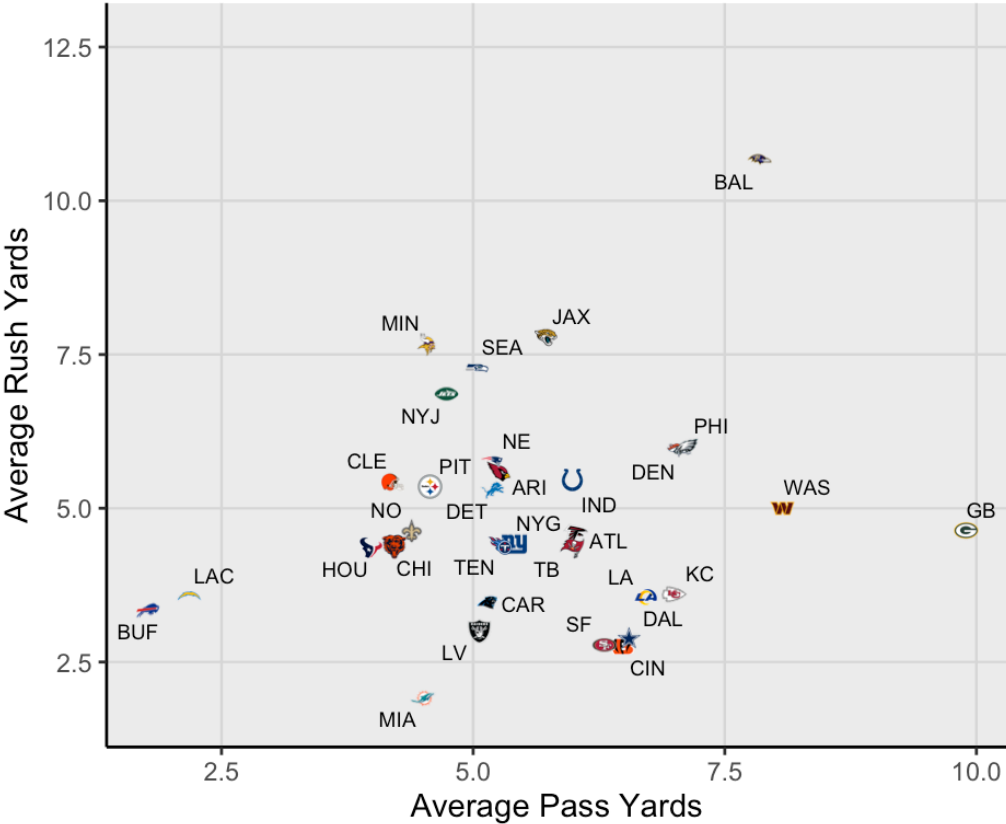
Average Rush vs. Pass Yards Per Play by Trailing Teams in Final 5 Minutes

2000 vs. 2024

2000 Season



2024 Season



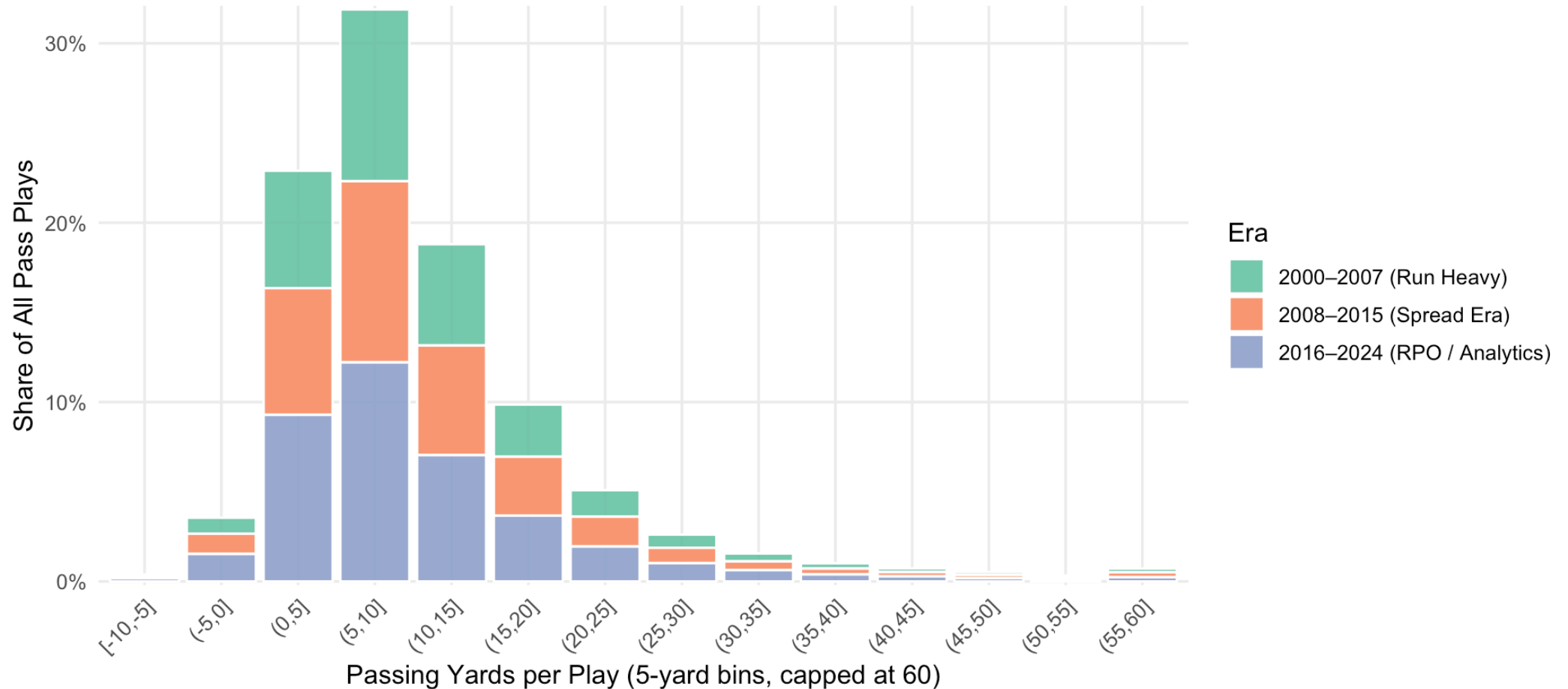
Passing Differentials in the Modern Era

- Recently, many offenses have resorted to shorter passes.
- Shorter passes are more consistent in comparison to longer, riskier passes.
- The sheer volume of passes have grown greatly between eras.
- Change in the pace of the game as well as rule changes

Passing Play Proportions Across Eras

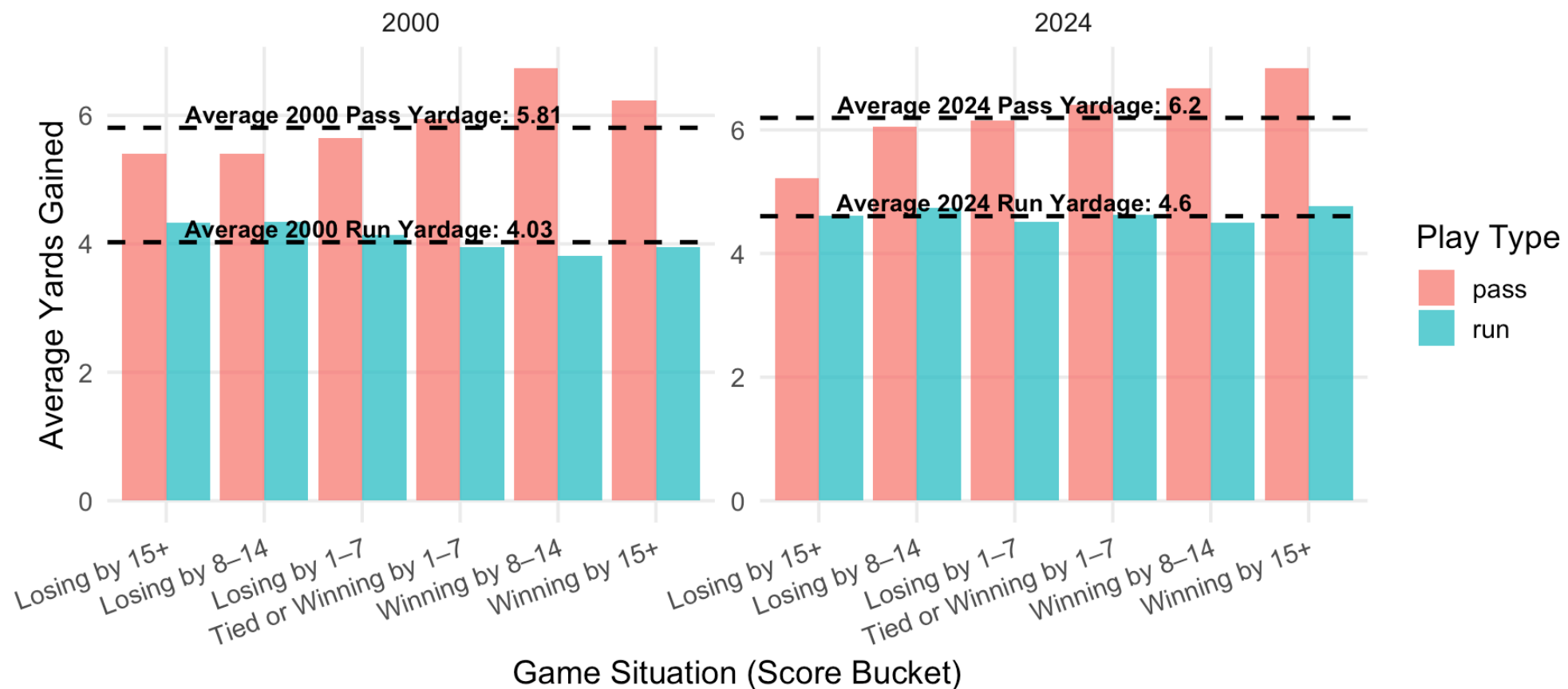
How Passing Play Outcomes Changed Across Eras (2000–2024)

Histogram of passing yards per play, with era volumes stacked



Changes in Pass vs. Run Yardage

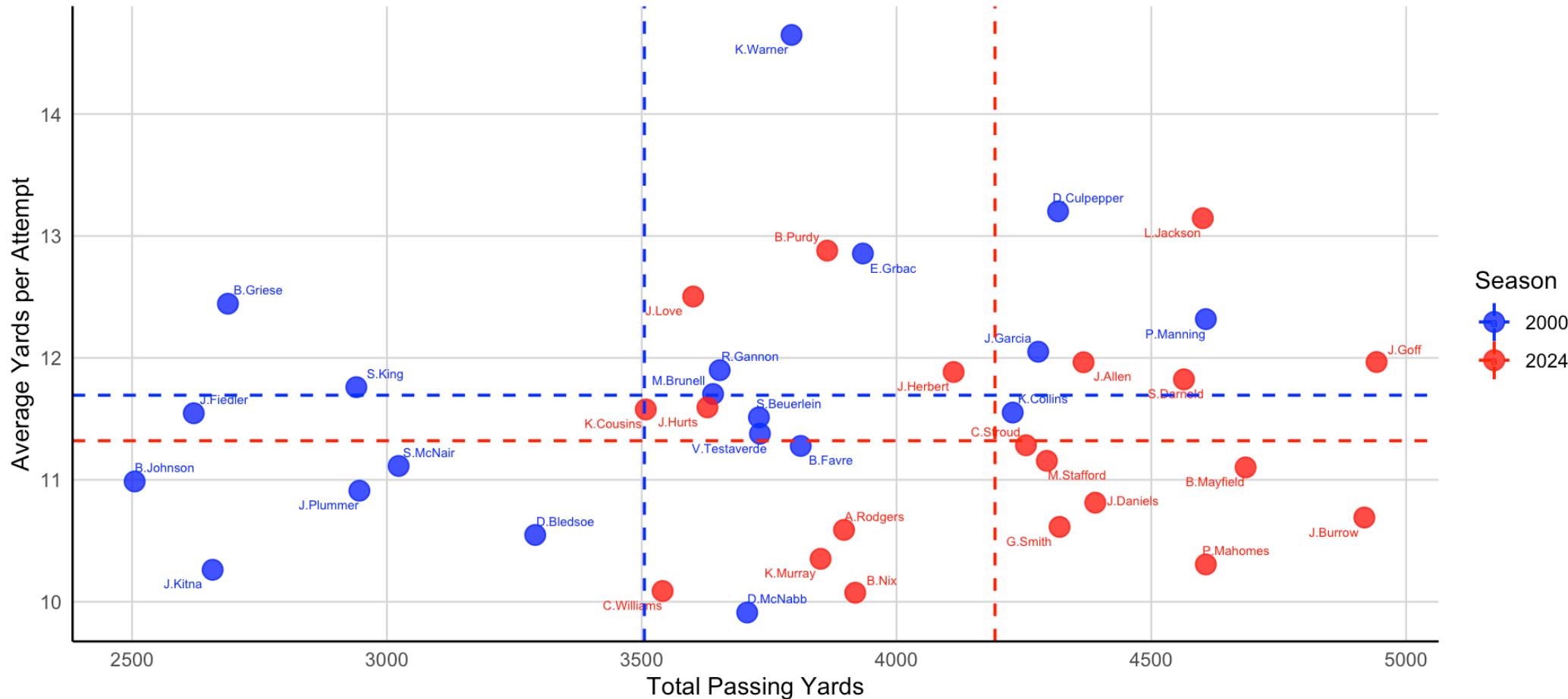
Average Yards Gained by Play Type Across Score Situations (2000 vs 2024)



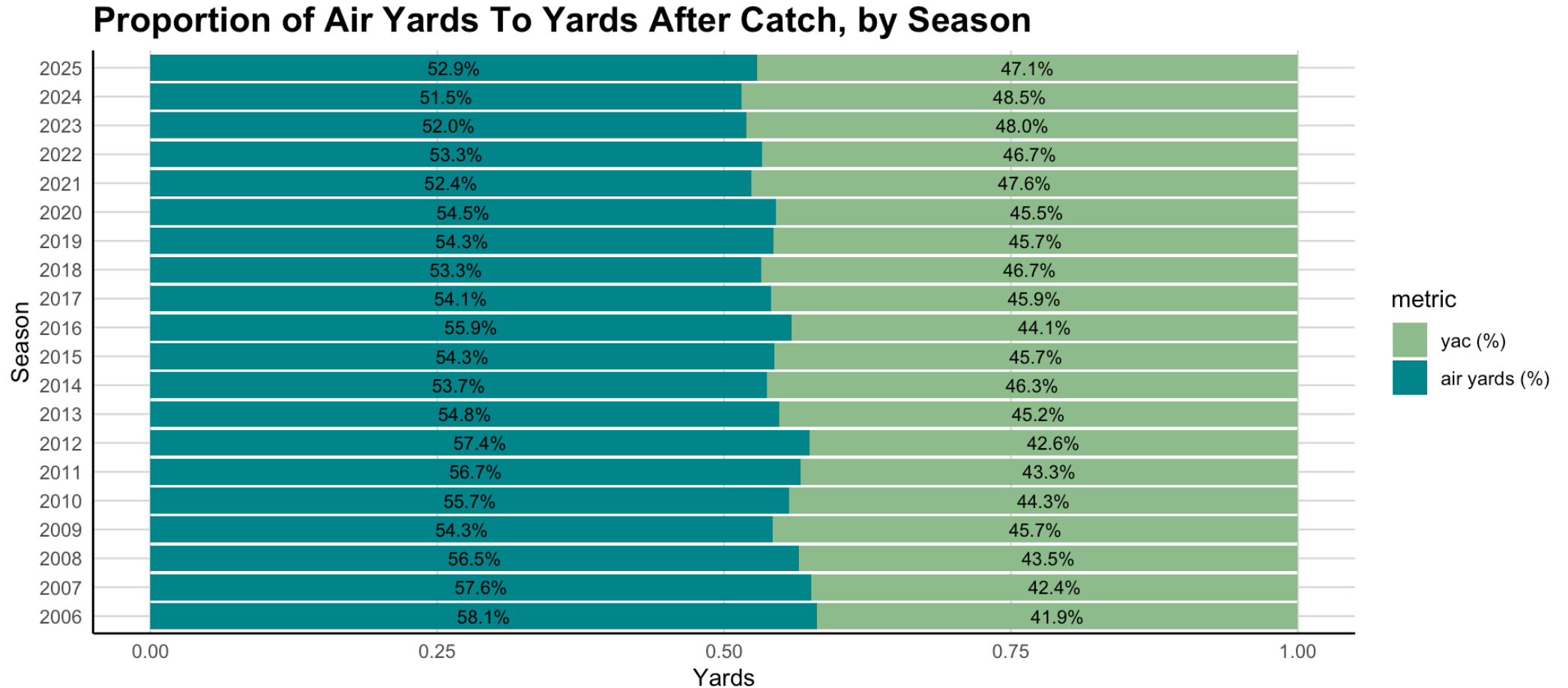
Pass Distance in Quarterbacks

QB Passing Yards: Total vs Average Yards (2000 vs 2024)

Top 20 Quarterbacks Per Season



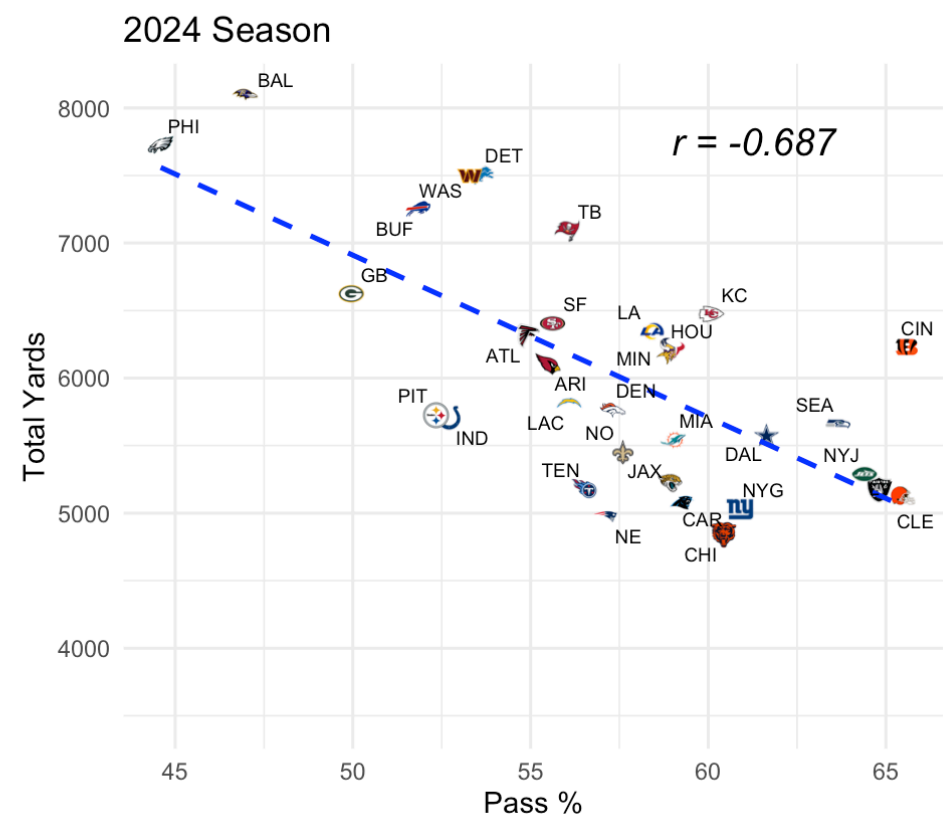
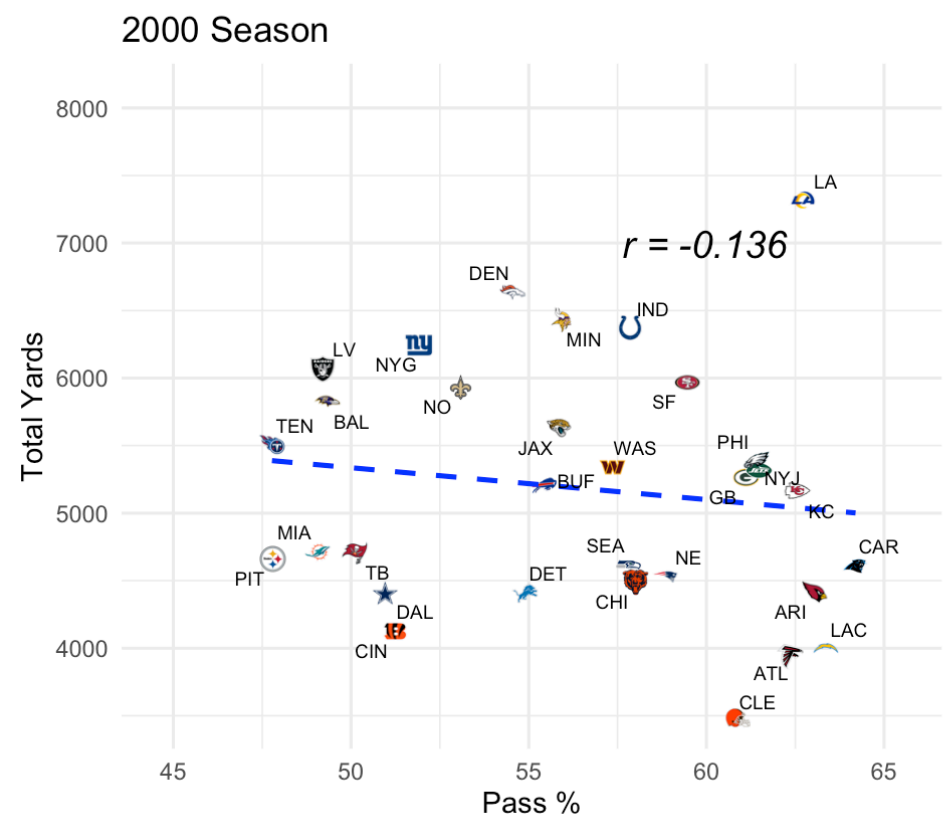
Proportionality of Air Yards to YAC



Trends in Passing Percentage to Offensive Yards

Passing Percentage vs. Offensive Yards Gained

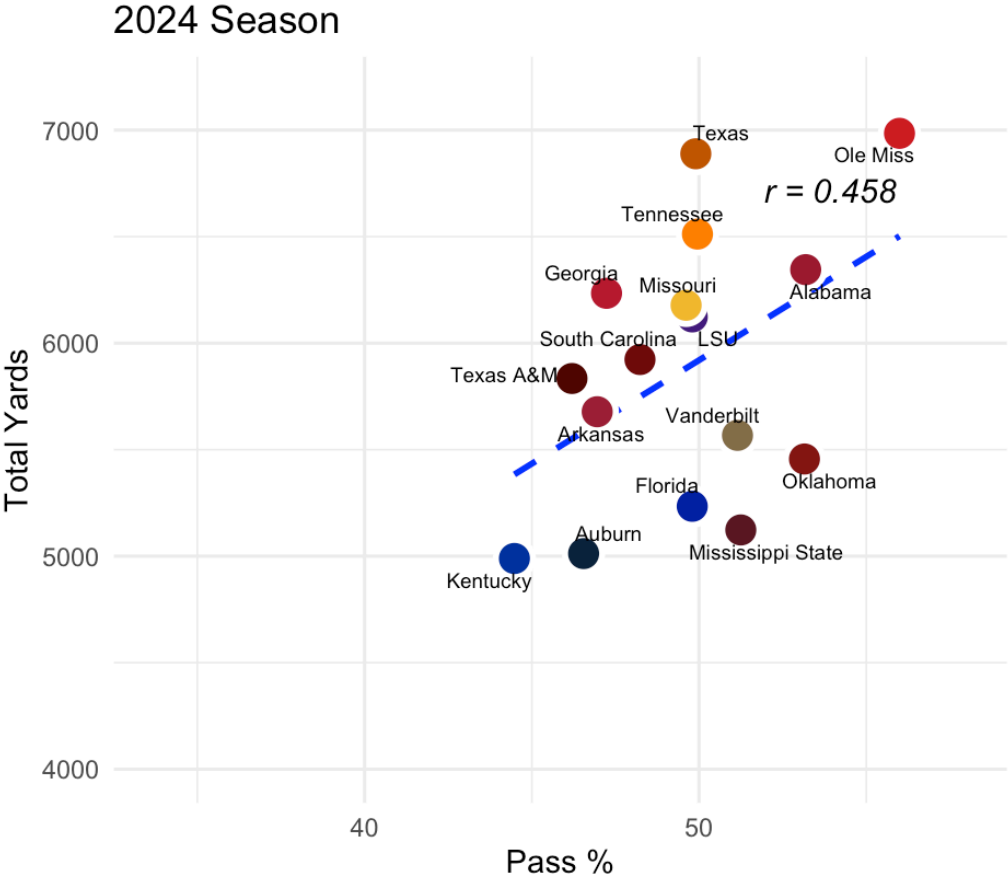
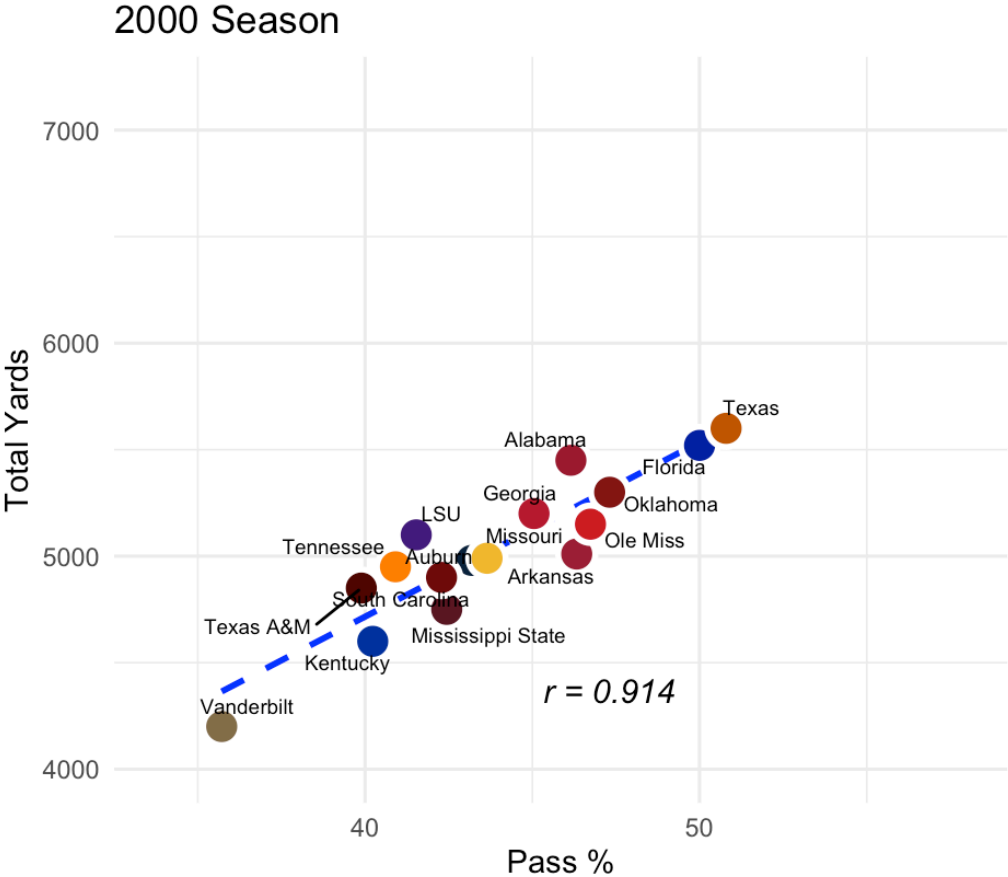
2000 vs 2024



Secondary Data: College Football

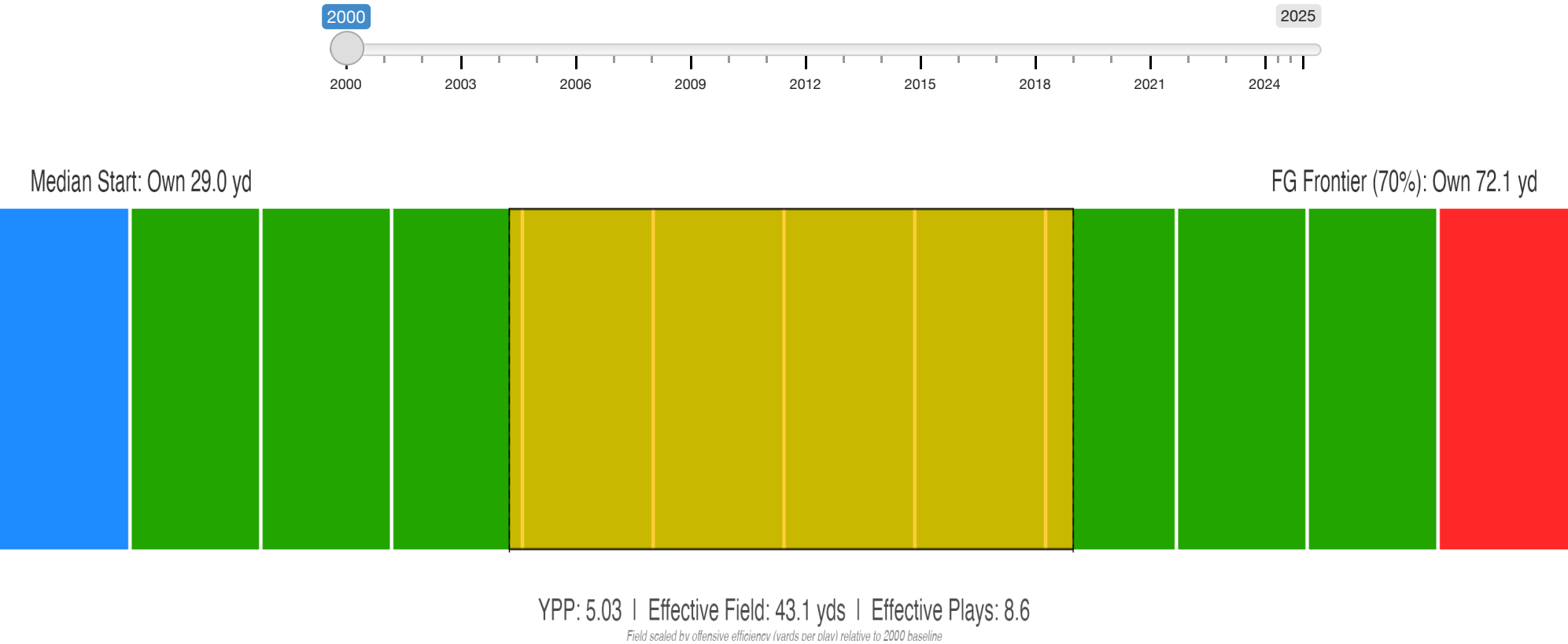
SEC Passing Percentage vs Total Offensive Yards

2000 vs 2024



Killer Plot

NFL Effective Field By Season



NFL Effective Field: Comparing Two Seasons

Season 1:

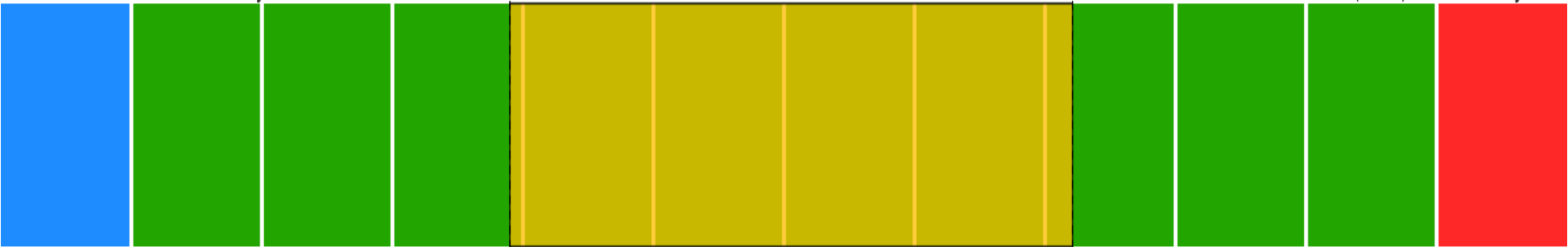
2000 ▾

Season 2:

2025 ▾

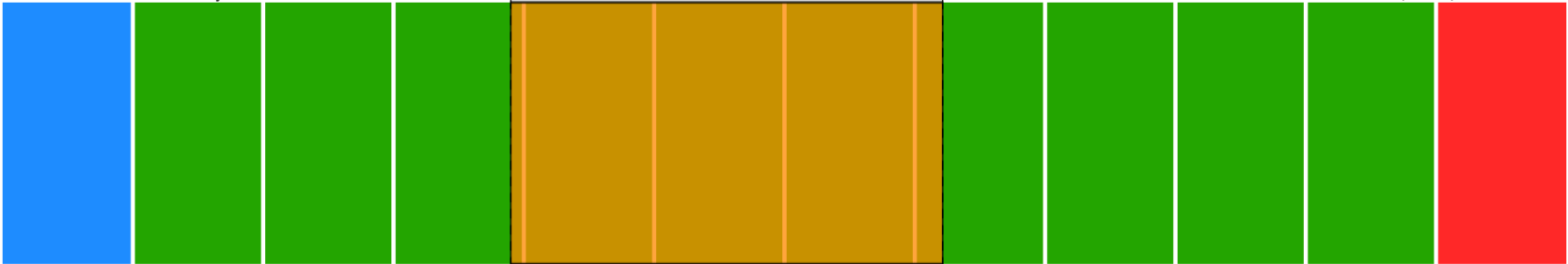
Median Start: Own 29.0 yd

FG Frontier (70%): Own 72.1 yd



Median Start: Own 29.0 yd

FG Frontier (70%): Own 62.1 yd



Takeaways

- Offenses are becoming more efficient as offensive strategy develops
- The passing game is becoming more diverse as offenses use different types of passing plays to exploit defensive strategy
- Some change occurs organically, however, the NFL still injects some regulated change into the game through their new kicking regulations and the new kickoff rule